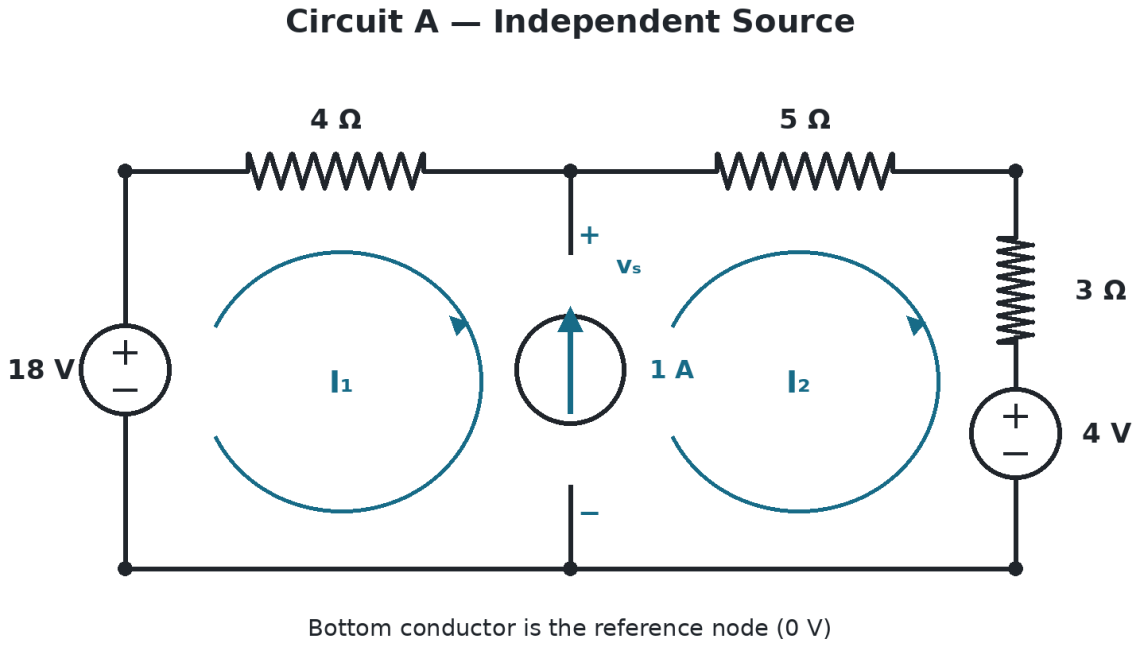
SuperMesh Questions

Question 1

Recognize the constraint. Write the current-source constraint and the outer-loop KVL equation. Do not solve.



Circuit A has clockwise mesh currents I_1 and I_2. The outer path contains a top-positive 18 V source, $4\ \Omega$ left resistor, $5\ \Omega$ top-right resistor, $3\ \Omega$ right resistor, and top-positive 4 V source. A shared 1 A source points upward. The bottom node is reference; v_s is positive at the shared source top.

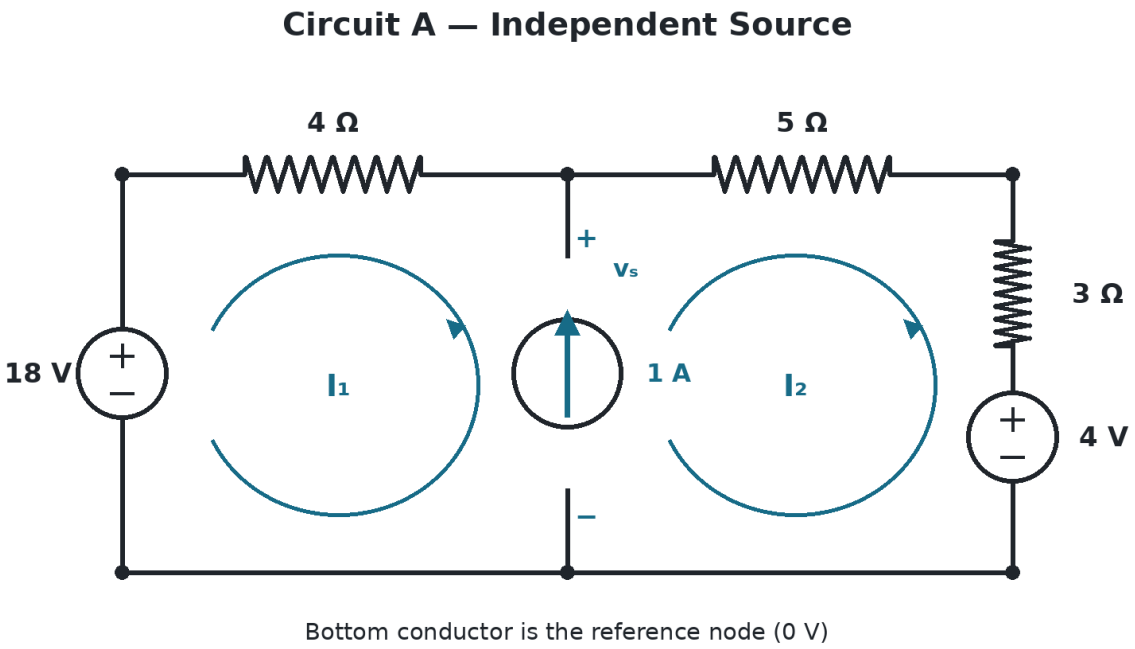
Solution

Let both mesh currents be clockwise and sum voltage drops around the supermesh. In the shared branch, I_2 is upward and I_1 is downward: $I_2 - I_1 = 1$

The outer-loop equation is $4I_1 + 5I_2 + 3I_2 + 4 - 18 = 0$; $4I_1 + 8I_2 = 14$

Question 2

Solve a basic supermesh. Find both mesh currents and state their directions.



Circuit A has clockwise mesh currents I_1 and I_2. The outer path contains a top-positive 18 V source, $4\ \Omega$ left resistor, $5\ \Omega$ top-right resistor, $3\ \Omega$ right resistor, and top-positive 4 V source. A shared 1 A source points upward. The bottom node is reference; v_s is positive at the shared source top.

Solution

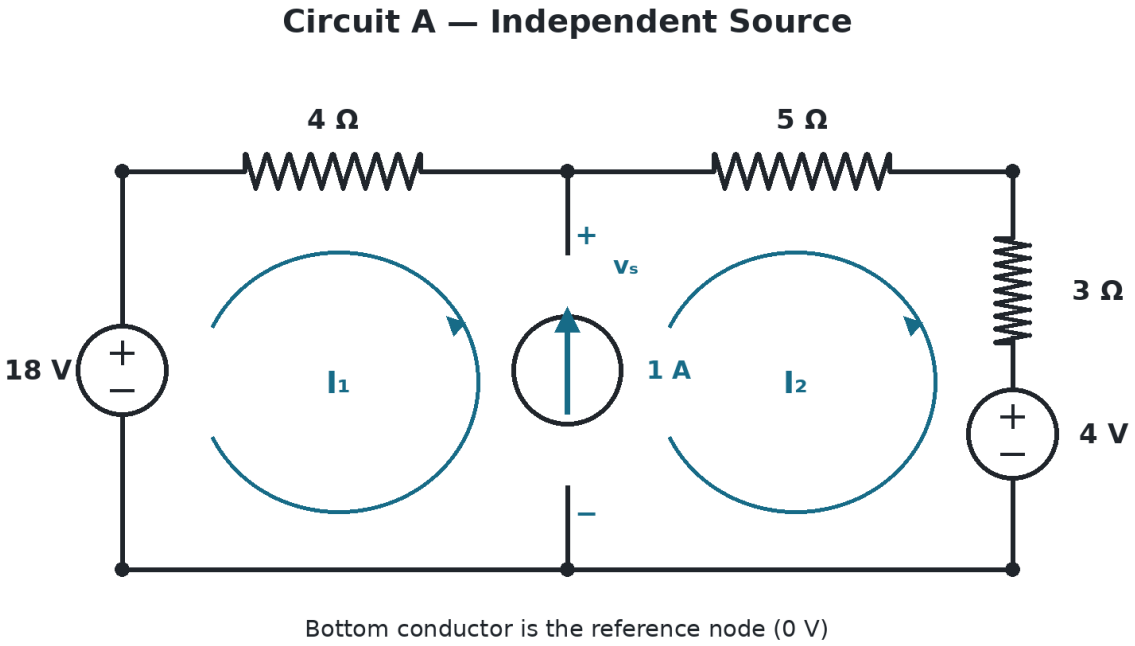
Use the equations $I_2 - I_1 = 1$; $4I_1 + 8I_2 = 14$

Substitute $I_2 = I_1 + 1$: $4I_1 + 8(I_1 + 1) = 14$; $I_1 = 0.500$ A, $I_2 = 1.50$ A

Both values are positive, so both currents are clockwise.

Question 3

Find the shared-source voltage. Find the top-to-bottom voltage v_s of the current source.



Circuit A has clockwise mesh currents I_1 and I_2. The outer path contains a top-positive 18 V source, $4\ \Omega$ left resistor, $5\ \Omega$ top-right resistor, $3\ \Omega$ right resistor, and top-positive 4 V source. A shared 1 A source points upward. The bottom node is reference; v_s is positive at the shared source top.

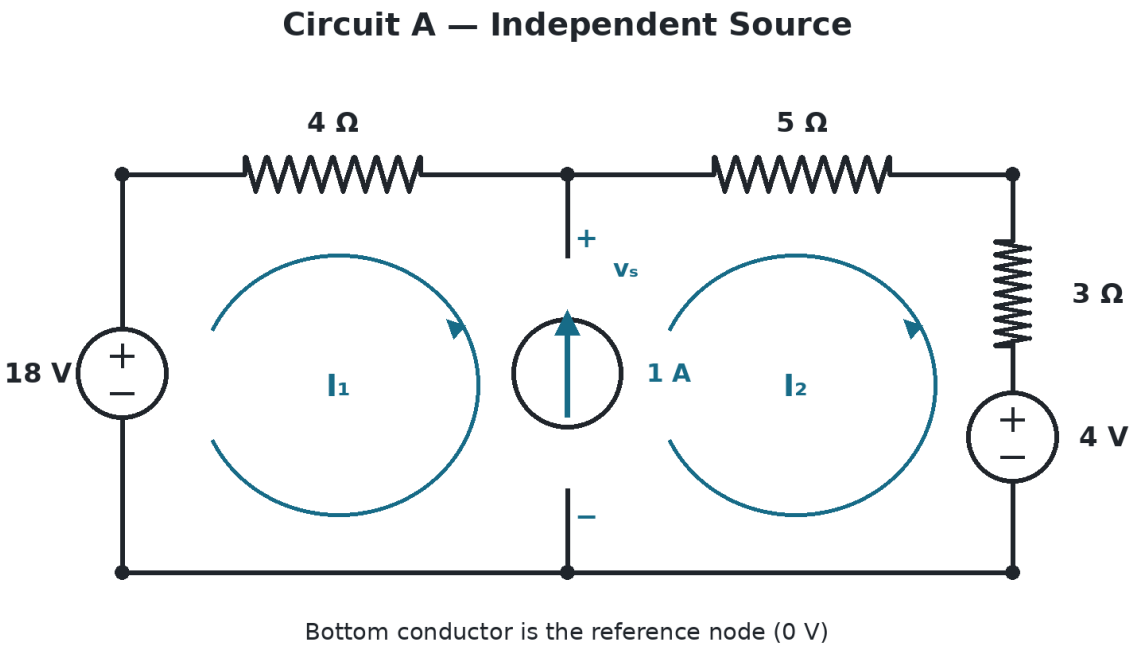
Solution

From Question 2, $I_1 = 0.500$ A. Traverse from reference through the left source and resistor: $v_s = 18 - 4I_1$; $v_s = 18 - 4(0.500) = 16.0$ V

The current-source top is 16.0 V above reference.

Question 4

Current-source power. Find the shared current source power and state absorbed or delivered.



Circuit A has clockwise mesh currents I_1 and I_2. The outer path contains a top-positive 18 V source, $4\ \Omega$ left resistor, $5\ \Omega$ top-right resistor, $3\ \Omega$ right resistor, and top-positive 4 V source. A shared 1 A source points upward. The bottom node is reference; v_s is positive at the shared source top.

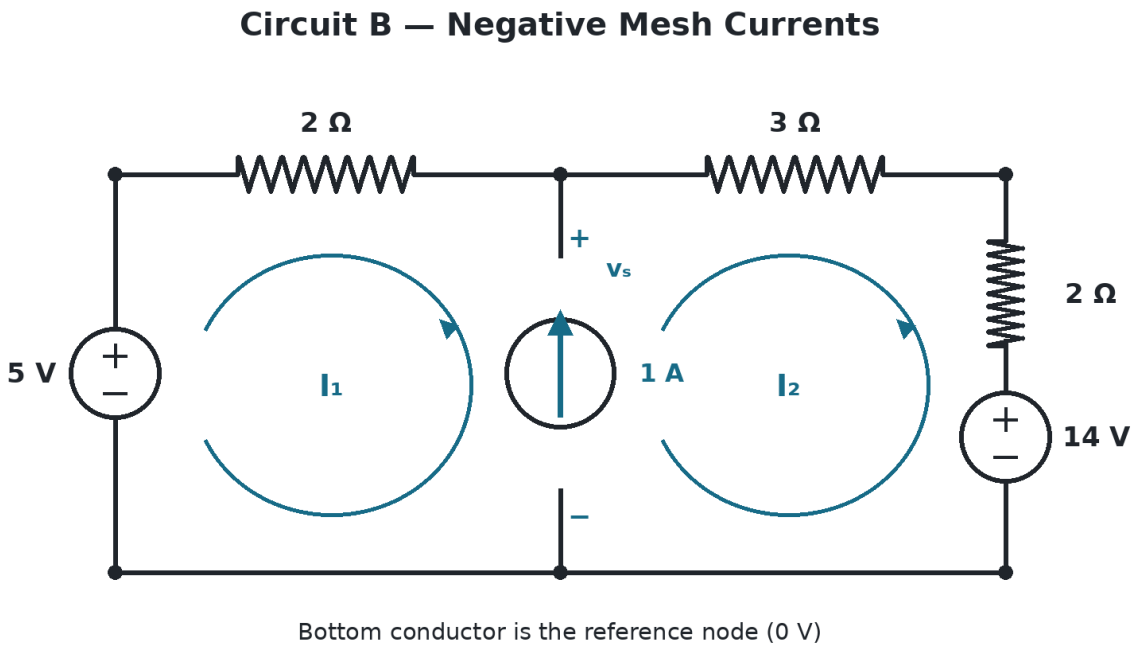
Solution

The source voltage is $v_s = 16.0$ V, positive at the top. Its 1 A current enters the negative terminal, so $p_s = -v_s(1) = -16.0$ W

The source delivers 16.0 W.

Question 5

Interpret negative mesh currents. Find both mesh currents and give their actual directions.



Circuit B has clockwise currents I_1 and I_2. The outer path contains top-positive sources of 5 V and 14 V, with $2\ \Omega$, $3\ \Omega$, and $2\ \Omega$ resistors. The shared 1 A source points upward. The bottom node is reference and v_s is top-positive.

Solution

For the upward source, $I_2 - I_1 = 1$

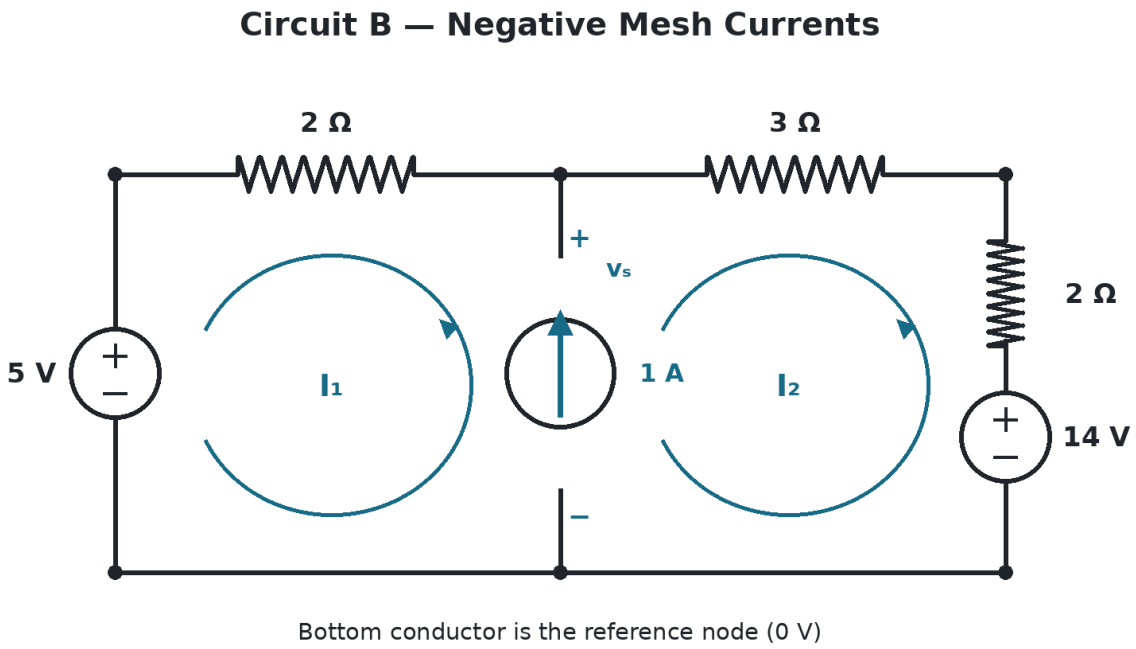
Outer-loop KVL gives $2I_1 + (3+2)I_2 = 5 - 14 = -9$

Substitute $I_2 = I_1 + 1$: $2I_1 + 5(I_1 + 1) = -9$; $I_1 = -2.00$ A, $I_2 = -1.00$ A

Both actual currents are counterclockwise: 2.00 A left and 1.00 A right.

Question 6

Power with reversed actual current. Find v_s and the shared-source power.



Circuit B has clockwise currents I_1 and I_2. The outer path contains top-positive sources of 5 V and 14 V, with $2\ \Omega$, $3\ \Omega$, and $2\ \Omega$ resistors. The shared 1 A source points upward. The bottom node is reference and v_s is top-positive.

Solution

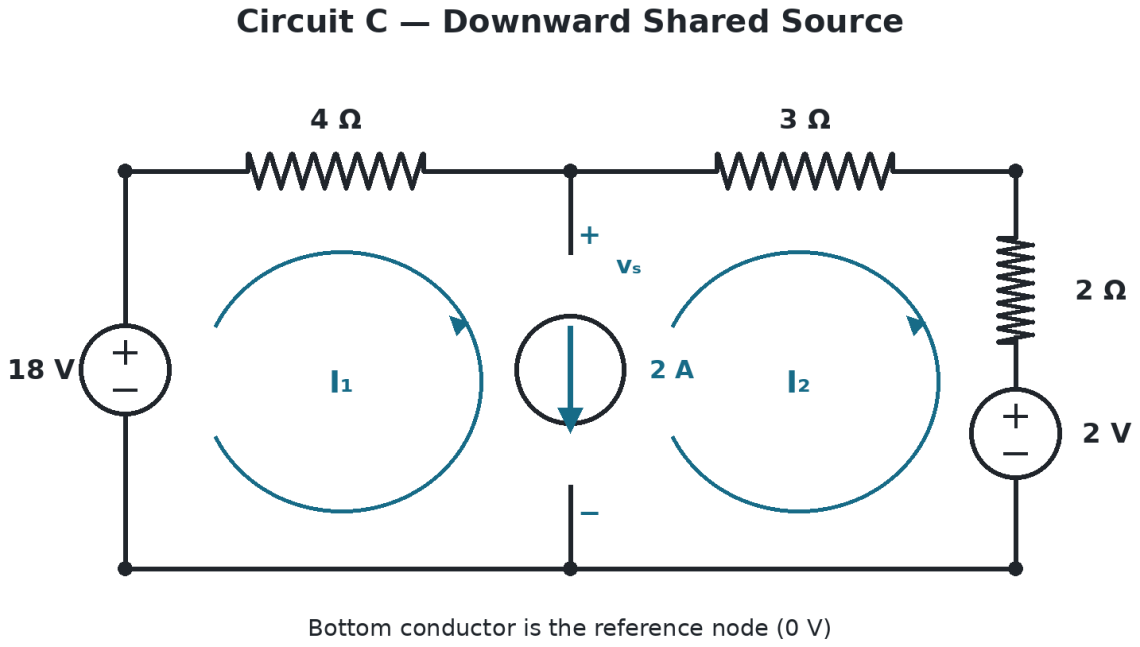
Using $I_1 = -2.00$ A, $v_s = 5 - 2I_1 = 5 - 2(-2) = 9.00$ V

The 1 A source current enters the negative terminal: $p_s = -v_s(1) = -9.00$ W

It delivers 9.00 W.

Question 7

Downward source constraint. Write the constraint, solve both currents, and state their directions.



Circuit C has clockwise currents I_1 and I_2. The outer path contains top-positive 18 V and 2 V sources with $4\ \Omega$, $3\ \Omega$, and $2\ \Omega$ resistors. The shared 2 A source points downward. The bottom node is reference and v_s is top-positive.

Solution

The downward source direction reverses the constraint: $I_1 - I_2 = 2$

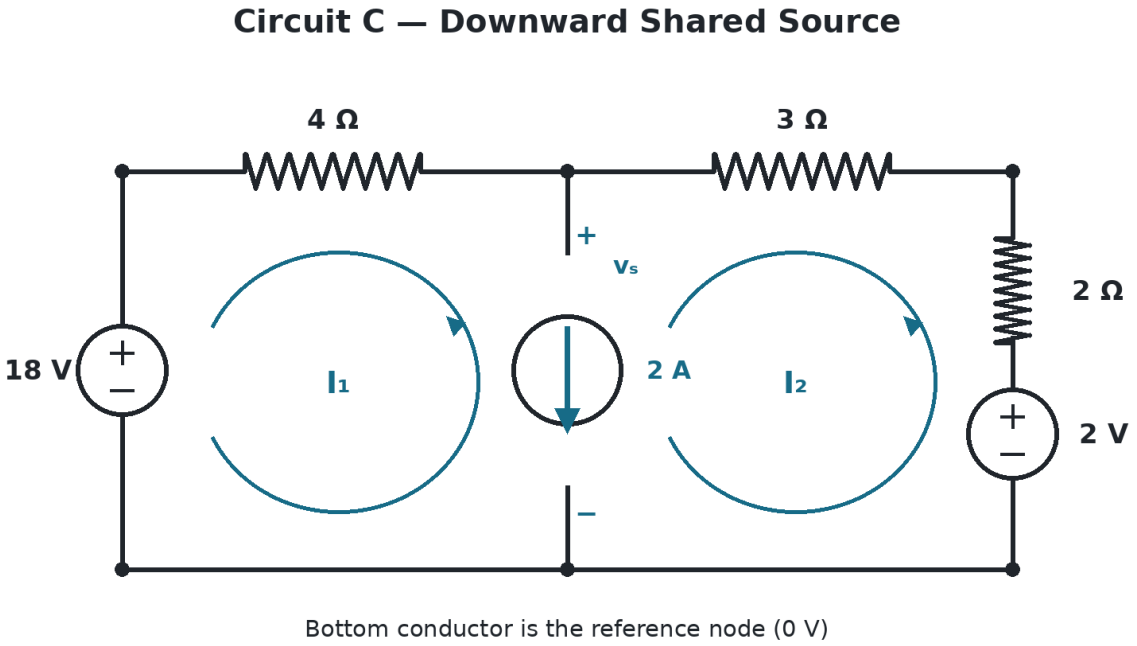
Outer-loop KVL is $4I_1 + (3+2)I_2 = 18 - 2 = 16$

With $I_1 = I_2 + 2$, $4(I_2 + 2) + 5I_2 = 16$; $I_2 = 0.889$ A, $I_1 = 2.89$ A

Both currents are clockwise.

Question 8

Power absorbed by a current source. Find the shared-source voltage and power.



Circuit C has clockwise currents I_1 and I_2. The outer path contains top-positive 18 V and 2 V sources with $4\ \Omega$, $3\ \Omega$, and $2\ \Omega$ resistors. The shared 2 A source points downward. The bottom node is reference and v_s is top-positive.

Solution

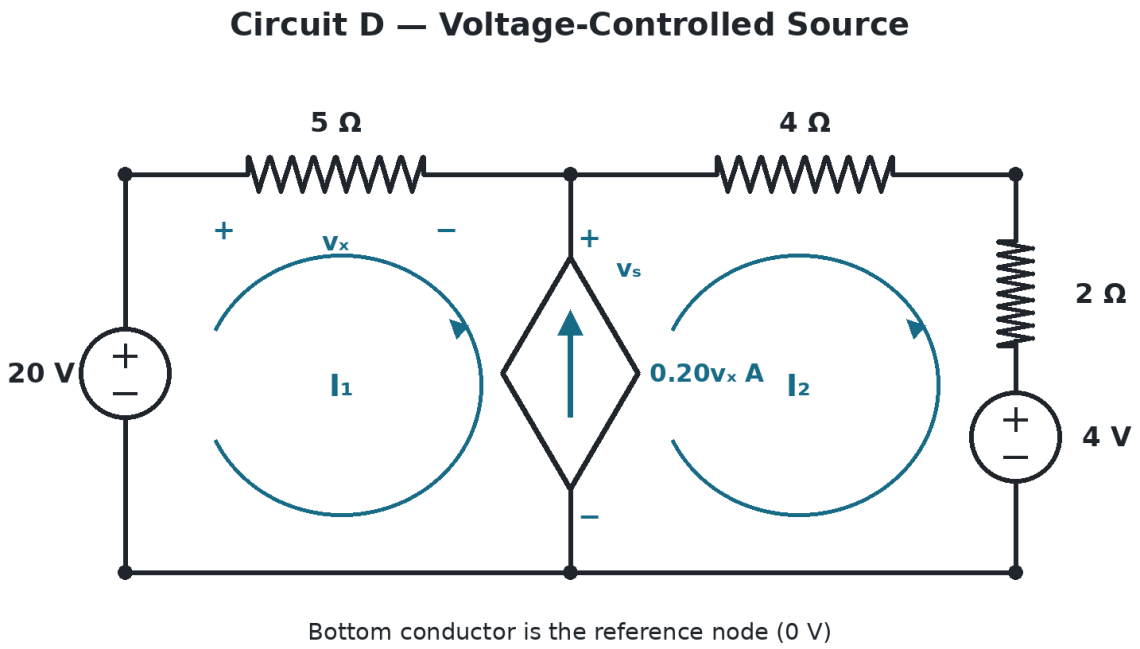
From the left path, $v_s = 18 - 4I_1 = 18 - 4(2.889) = 6.44$ V

The 2 A downward current enters the positive terminal: $p_s = v_s(2) = 12.9$ W

The current source absorbs 12.9 W.

Question 9

Voltage-controlled source setup. Express the source constraint only in terms of I_1 and I_2.



Circuit D has clockwise currents I_1 and I_2. Its outer path contains top-positive 20 V and 4 V sources and $5\,\Omega$, $4\,\Omega$, and $2\,\Omega$ resistors. The shared dependent source points upward and equals $0.20v_x$ A; v_x is positive left-to-right across the $5\,\Omega$ resistor. The bottom node is reference.

Solution

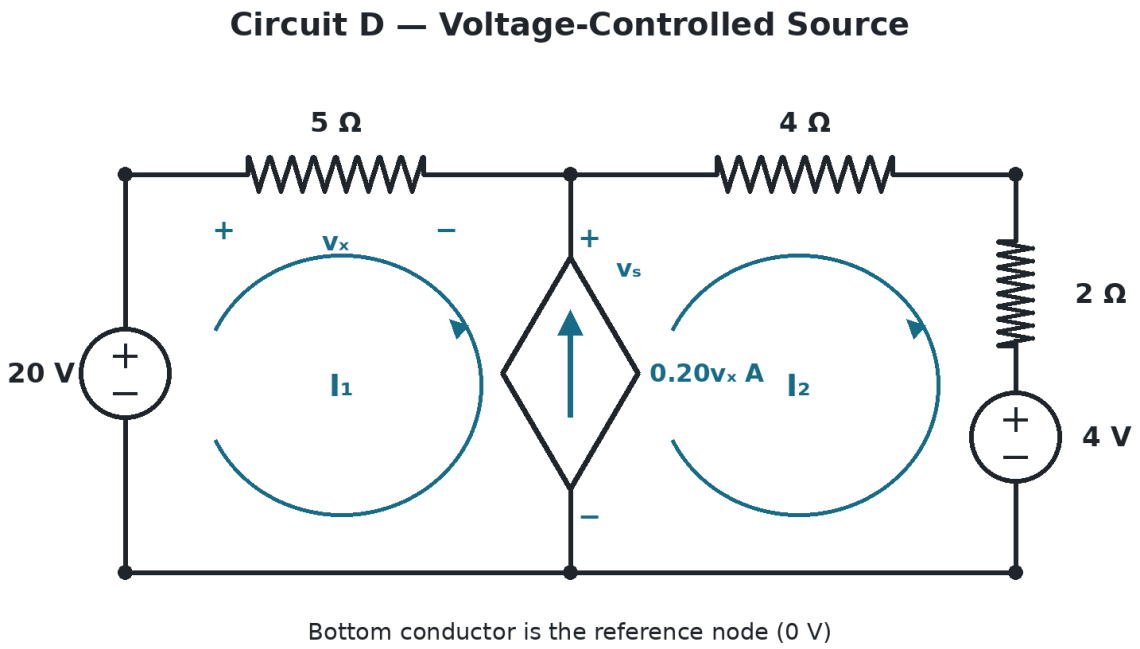
The controlling voltage uses the marked resistor polarity: $v_x = 5I_1$

The upward shared-source constraint is $I_2 - I_1 = 0.20v_x$

Therefore, $I_2 - I_1 = 0.20(5I_1) = I_1$; $I_2 = 2I_1$

Question 10

Solve the dependent-source supermesh. Find both mesh currents.



Circuit D has clockwise currents I_1 and I_2. Its outer path contains top-positive 20 V and 4 V sources and $5\,\Omega$, $4\,\Omega$, and $2\,\Omega$ resistors. The shared dependent source points upward and equals $0.20v_x$ A; v_x is positive left-to-right across the $5\,\Omega$ resistor. The bottom node is reference.

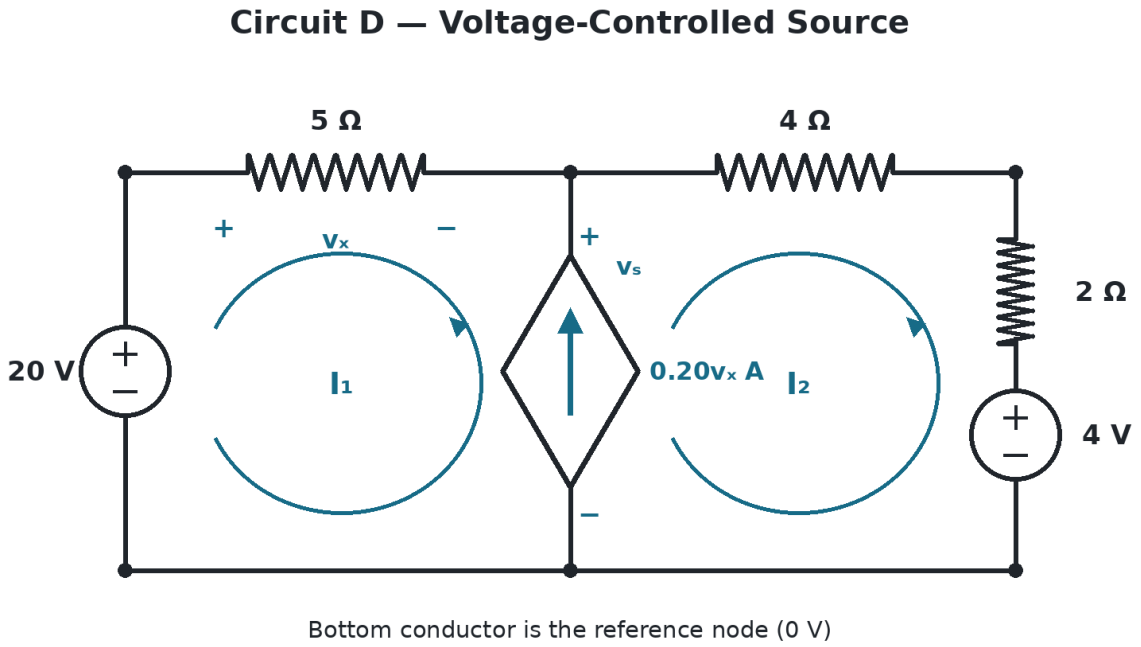
Solution

Use $I_2 = 2I_1$. Outer-loop KVL gives $5I_1 + (4+2)I_2 = 20 - 4 = 16$; $5I_1 + 6(2I_1) = 16$; $I_1 = 0.941$ A, $I_2 = 1.88$ A

Both currents are clockwise.

Question 11

Dependent-source voltage and power. Find the shared-source voltage and state whether it absorbs or delivers power.



Circuit D has clockwise currents I_1 and I_2. Its outer path contains top-positive 20 V and 4 V sources and $5\,\Omega$, $4\,\Omega$, and $2\,\Omega$ resistors. The shared dependent source points upward and equals $0.20v_x$ A; v_x is positive left-to-right across the $5\,\Omega$ resistor. The bottom node is reference.

Solution

The source voltage is $v_s = 20 - 5I_1 = 20 - 5(0.941) = 15.3$ V

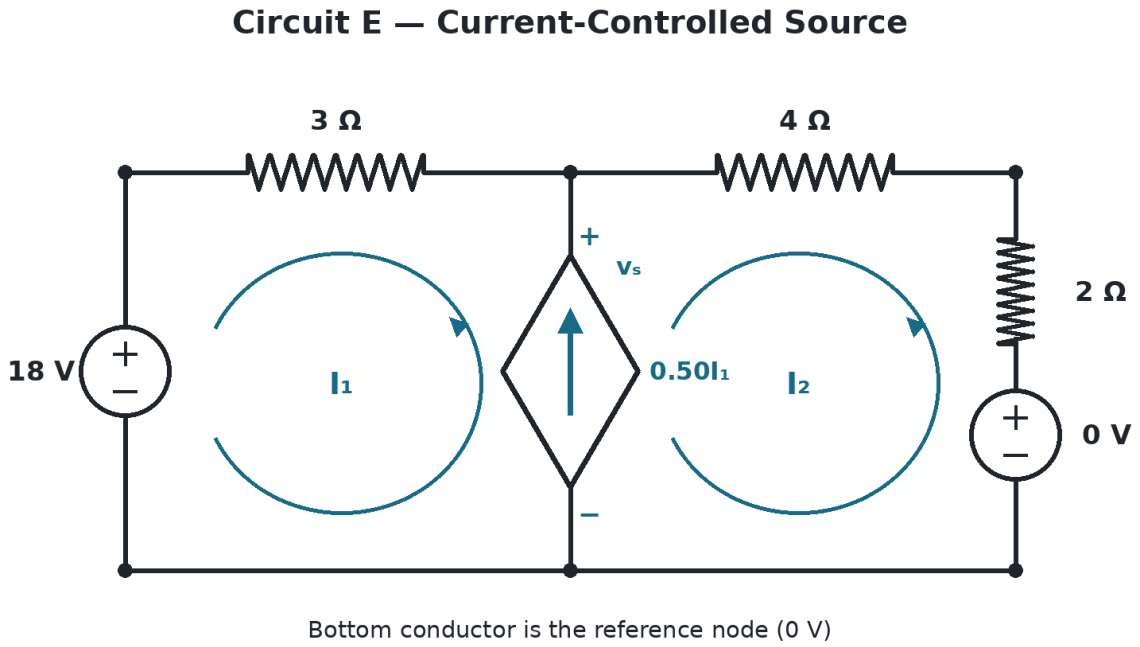
Its upward current is $0.20v_x = I_2 - I_1 = 0.941$ A

The current enters the negative terminal: $p_s = -v_s(0.941) = -14.4$ W

The dependent source delivers 14.4 W.

Question 12

Current-controlled source. Find both mesh currents.



Circuit E has clockwise currents I_1 and I_2. Its outer path contains a top-positive 18 V source, a zero-volt source, and $3\ \Omega$, $4\ \Omega$, and $2\ \Omega$ resistors. The shared dependent source points upward and equals $0.50I_1$. The bottom node is reference and v_s is top-positive.

Solution

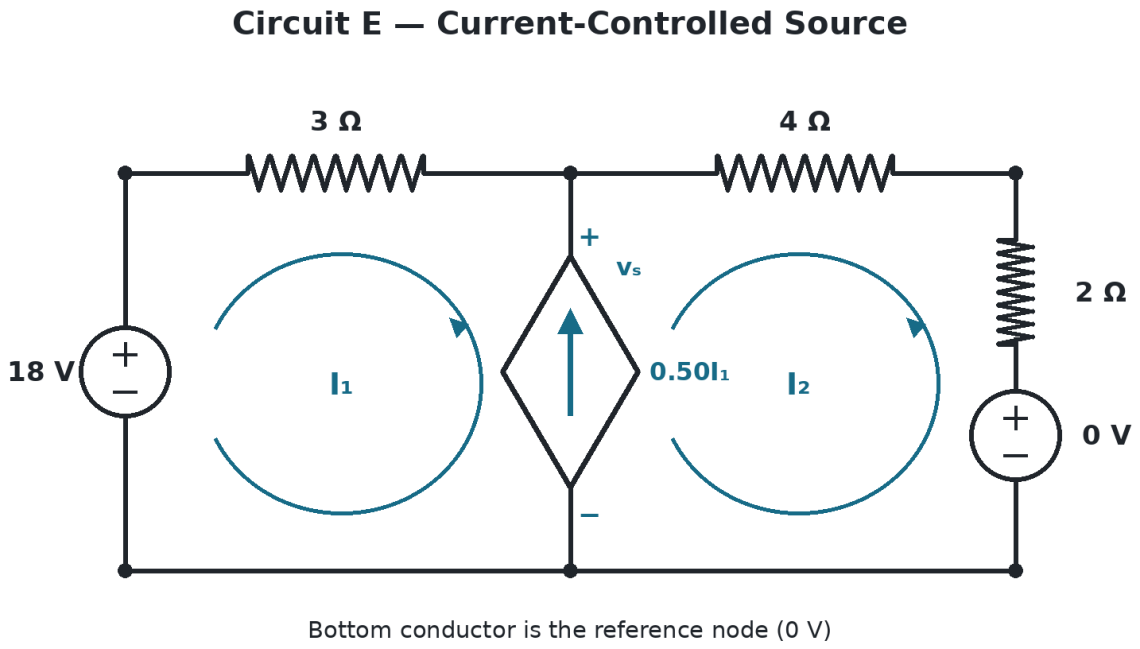
The source constraint is $I_2 - I_1 = 0.50I_1$; $I_2 = 1.50I_1$

Outer-loop KVL gives $3I_1 + (4+2)I_2 = 18$; $3I_1 + 6(1.50I_1) = 18$; $I_1 = 1.50$ A, $I_2 = 2.25$ A

Both currents are clockwise.

Question 13

Check a dependent source value. Find the actual shared-branch current and verify it from the mesh currents.



Circuit E has clockwise currents I_1 and I_2. Its outer path contains a top-positive 18 V source, a zero-volt source, and $3\ \Omega$, $4\ \Omega$, and $2\ \Omega$ resistors. The shared dependent source points upward and equals $0.50I_1$. The bottom node is reference and v_s is top-positive.

Solution

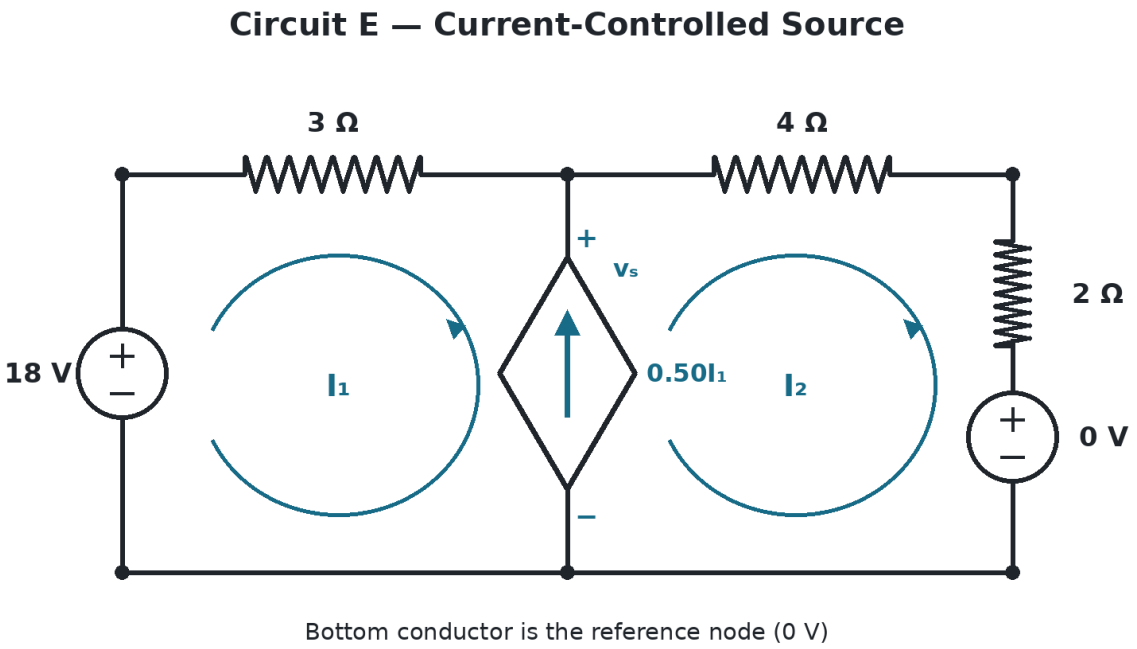
From the controlling relation, $i_s = 0.50I_1 = 0.50(1.50) = 0.750$ A upward

From the mesh currents, $i_s = I_2 - I_1 = 2.25 - 1.50 = 0.750$ A upward

The two evaluations agree.

Question 14

Dependent-source power. Find v_s and the power of the dependent source.



Circuit E has clockwise currents I_1 and I_2. Its outer path contains a top-positive 18 V source, a zero-volt source, and $3\ \Omega$, $4\ \Omega$, and $2\ \Omega$ resistors. The shared dependent source points upward and equals $0.50I_1$. The bottom node is reference and v_s is top-positive.

Solution

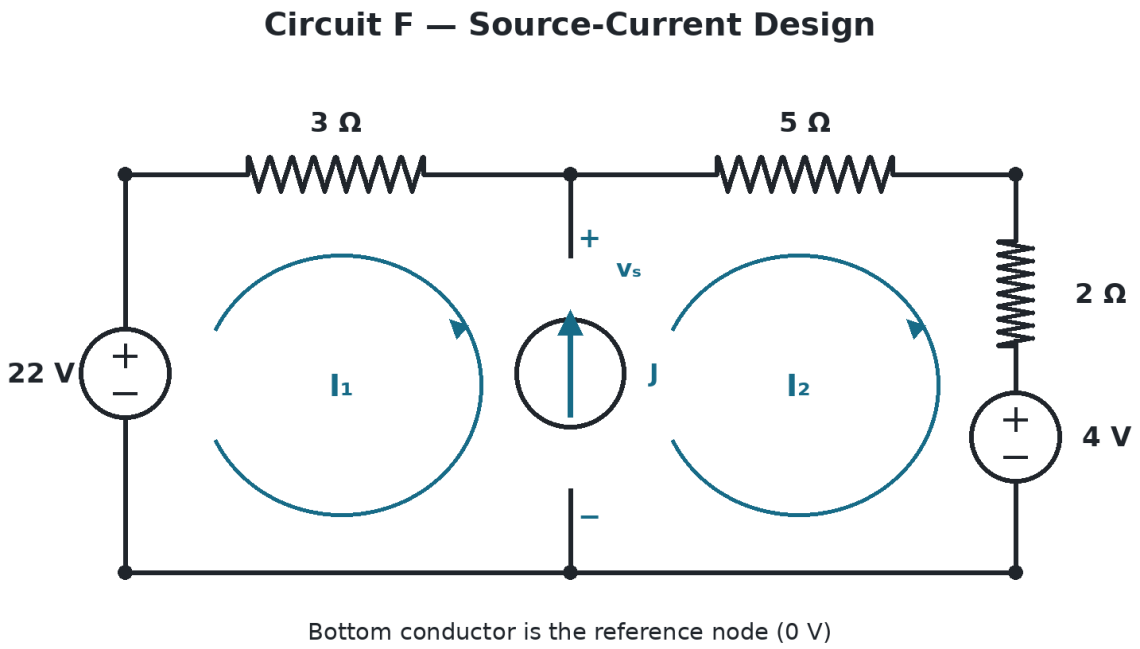
From the left path, $v_s = 18 - 3I_1 = 18 - 3(1.50) = 13.5$ V

The upward current enters the negative terminal: $p_s = -v_s(0.750) = -(13.5)(0.750) = -10.1$ W

The dependent source delivers 10.1 W.

Question 15

Design a shared current source. Choose J so that $I_2 = 3.00\ extA$ clockwise. Find I_1 and J.



Bottom conductor is the reference node (0 V)

Circuit F has clockwise currents I_1 and I_2. Its outer path contains top-positive 22 V and 4 V sources and $3\ \Omega$, $5\ \Omega$, and $2\ \Omega$ resistors. The shared unknown source J points upward. The bottom node is reference and v_s is top-positive.

Solution

Outer-loop KVL is $3I_1 + (5+2)I_2 = 22 - 4 = 18$

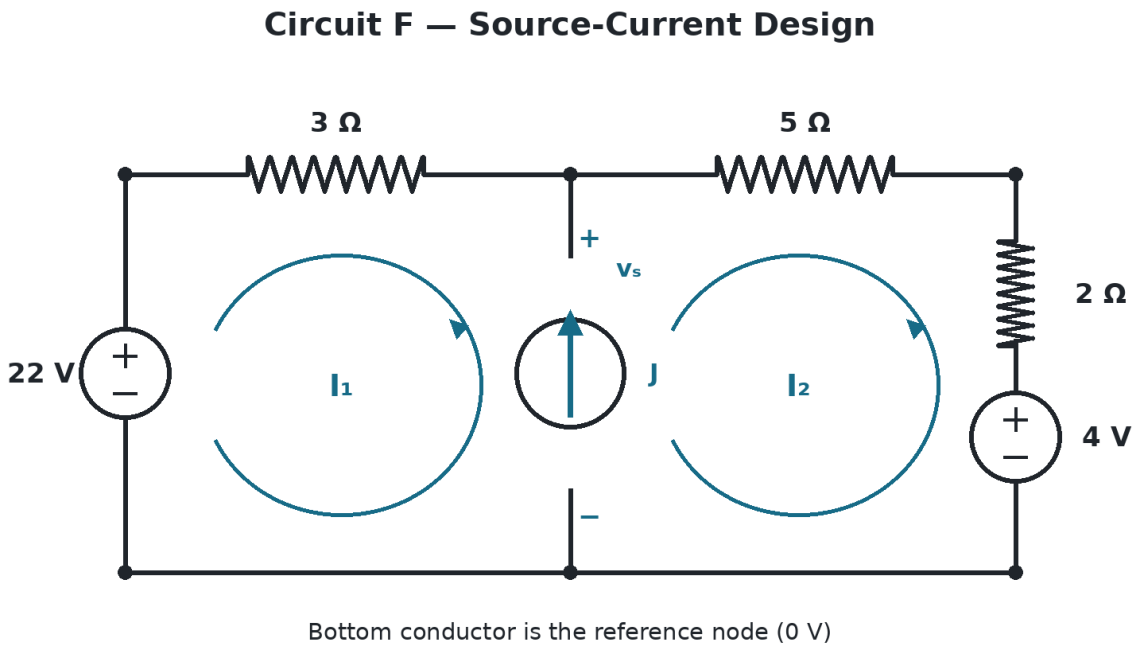
Substitute the target current: $3I_1 + 7(3.00) = 18$; $I_1 = -1.00$ A

For an upward shared source, $J = I_2 - I_1 = 3.00 - (-1.00) = 4.00$ A

The left current is actually 1.00 A counterclockwise.

Question 16

Power of a designed source. For the result of Question 15, find the shared source power.



Circuit F has clockwise currents I_1 and I_2. Its outer path contains top-positive 22 V and 4 V sources and $3\ \Omega$, $5\ \Omega$, and $2\ \Omega$ resistors. The shared unknown source J points upward. The bottom node is reference and v_s is top-positive.

Solution

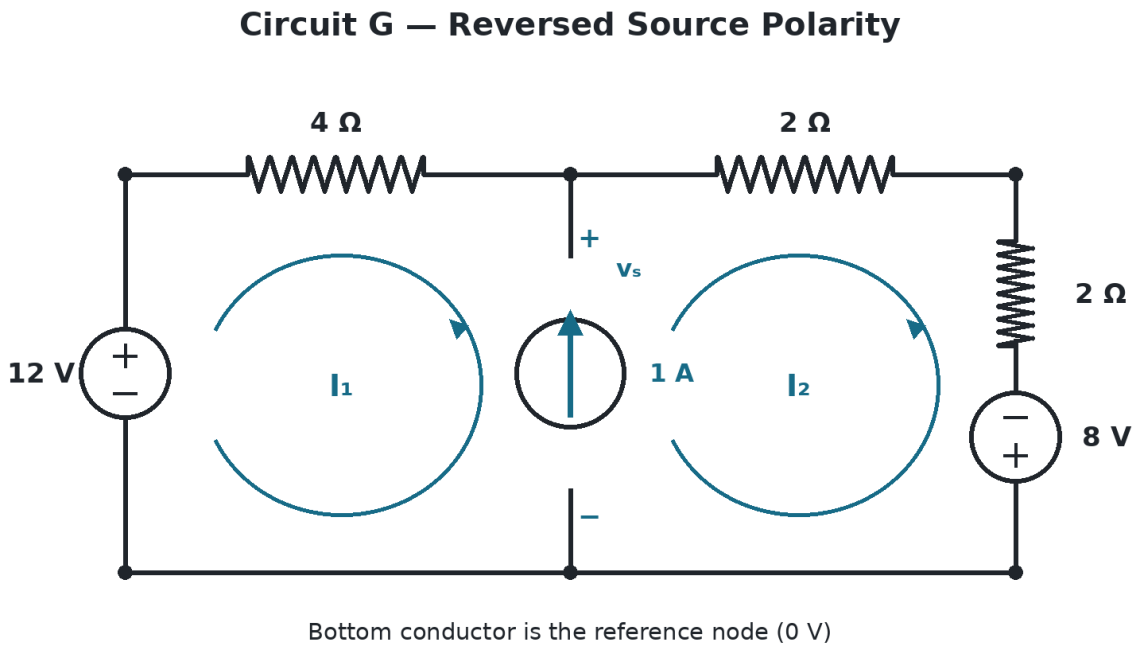
The top-to-bottom source voltage is $v_s = 22 - 3I_1 = 22 - 3(-1.00) = 25.0$ V

The 4.00 A current points upward and enters the negative terminal: $p_s = -v_s J = -(25.0)(4.00) = -100$ W

The source delivers 100 W.

Question 17

Reversed voltage-source polarity. Find both mesh currents. Account for the right source polarity.



Circuit G has clockwise currents I_1 and I_2. Its outer path contains a top-positive 12 V source, $4\ \Omega$, $2\ \Omega$, and $2\ \Omega$ resistors, then an 8 V source positive at the bottom. The shared 1 A source points upward. The bottom node is reference.

Solution

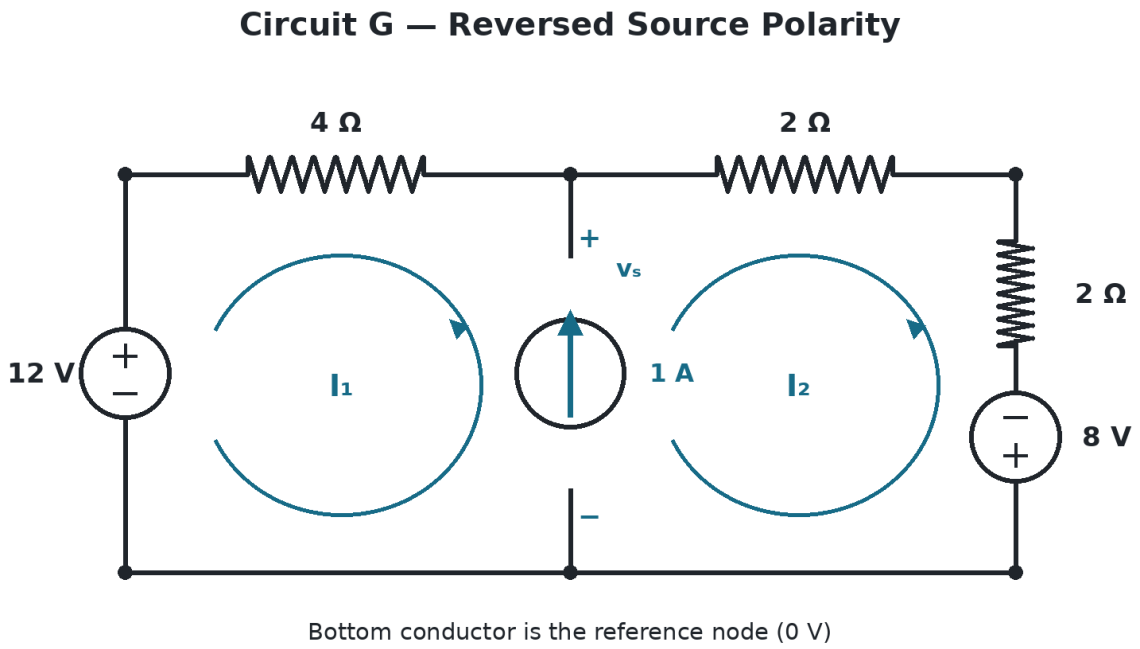
The right source is a rise when the outer loop is traversed downward. Thus, $I_2 - I_1 = 1$; $4I_1 + (2 + 2)I_2 = 12 + 8 = 20$

With $I_2 = I_1 + 1$, $4I_1 + 4(I_1 + 1) = 20$; $I_1 = 2.00$ A, $\quad I_2 = 3.00$ A

Both currents are clockwise.

Question 18

Resistor power distribution. Find the power absorbed by each resistor and their total.



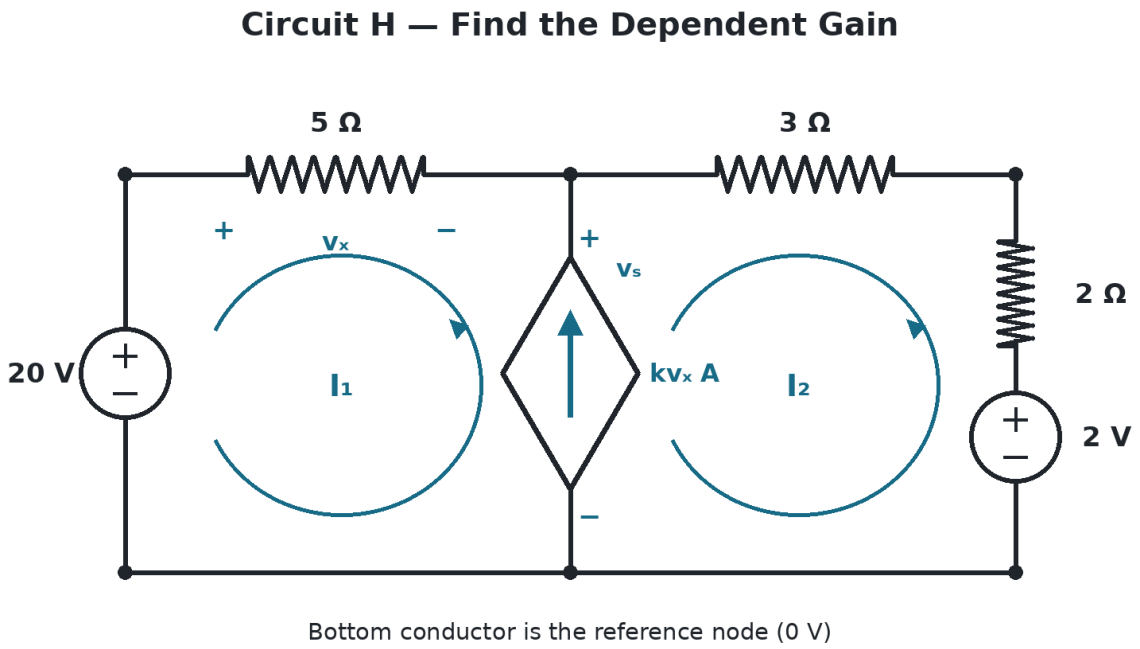
Circuit G has clockwise currents I_1 and I_2. Its outer path contains a top-positive 12 V source, $4\ \Omega$, $2\ \Omega$, and $2\ \Omega$ resistors, then an 8 V source positive at the bottom. The shared 1 A source points upward. The bottom node is reference.

Solution

Using $I_1 = 2.00$ A and $I_2 = 3.00$ A, $P_{4\Omega} = I_1^2(4) = (2.00)^2(4) = 16.0$ W; $P_{2\Omega,\text{top}} = I_2^2(2) = (3.00)^2(2) = 18.0$ W; $P_{2\Omega,\text{right}} = I_2^2(2) = 18.0$ W; $P_R = 16.0 + 18.0 + 18.0 = 52.0$ W absorbed

Question 19

Determine a dependent gain. A meter reports $I_1 = 1.00\ extA$ clockwise. Find I_2 and the gain k.



Circuit H has clockwise currents I_1 and I_2. Its outer path contains top-positive 20 V and 2 V sources and $5\ \Omega$, $3\ \Omega$, and $2\ \Omega$ resistors. The shared dependent source kv_x points upward; v_x is positive left-to-right across the $5\ \Omega$ resistor. The bottom node is reference.

Solution

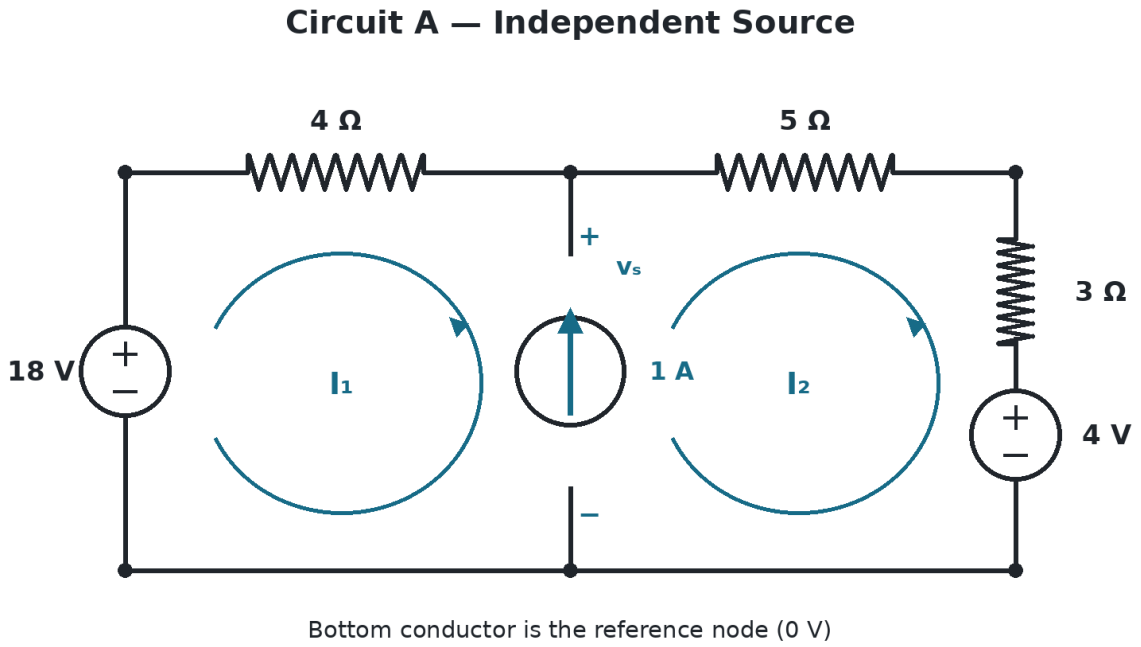
Outer-loop KVL gives $5I_1 + (3+2)I_2 = 20 - 2 = 18$; $5(1.00) + 5I_2 = 18$; $I_2 = 2.60$ A

The control voltage and source current are $v_x = 5I_1 = 5.00$ V; $i_s = I_2 - I_1 = 2.60 - 1.00 = 1.60$ A

Since $i_s = kv_x$, $k = \frac{1.60}{5.00} = 0.320$ A/V $= 0.320$ S

Question 20

Power-balance audit. Verify conservation of power for every element. Use positive power for absorption.



Circuit A has clockwise mesh currents I_1 and I_2. The outer path contains a top-positive 18 V source, $4\ \Omega$ left resistor, $5\ \Omega$ top-right resistor, $3\ \Omega$ right resistor, and top-positive 4 V source. A shared 1 A source points upward. The bottom node is reference; v_s is positive at the shared source top.

Solution

From Questions 2 and 3, $I_1 = 0.500$ A, $I_2 = 1.50$ A, and $v_s = 16.0$ V. Resistor absorption is $P_R = 4I_1^2 + 5I_2^2 + 3I_2^2$; $P_R = 4(0.500)^2 + 8(1.50)^2 = 19.0$ W

The left source current leaves its positive terminal, the right source current enters its positive terminal, and the current-source current enters its negative terminal: $p_{18\text{V}} = -18I_1 = -9.00$ W; $p_{4\text{V}} = 4I_2 = 6.00$ W; $p_s = -v_s(1) = -16.0$ W

Therefore, $\sum p = 19.0 - 9.00 + 6.00 - 16.0 = 0$

Absorbed and delivered power balance.